Association of Sleep Duration with Mortality from Cardiovascular Disease and Other Causes for Japanese Men and Women: the JACC Study

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Study Objectives: To examine sex-specific associations between sleep duration and mortality from cardiovascular disease and other causes. Design: Cohort study. Setting: Community-based study. Participants: A total of 98,634 subjects (41,489 men and 57,145 women) aged 40 to 79 years from 1988 to 1990 and were followed until 2003. Interventions: N/A. Measurements and Results: During a median follow-up of 14.3 years, there were 1964 deaths (men and women: 1038 and 926) from stroke, 881 (508 and 373) from coronary heart disease, 4287 (2297 and 1990) from cardiovascular disease, 5465 (3432 and 2033) from cancer, and 14,540 (8548 and 5992) from all causes. Compared with a sleep duration of 7 hours, sleep duration of 4 hours or less was associated with increased mortality from coronary heart disease for women and noncardiovascular disease/noncancer and all causes in both sexes. The respective multivariable hazard ratios were 2.32 (1.19–4.50) for coronary heart disease in women, 1.49 (1.02–2.18) and 1.47 (1.01–2.15) for noncardiovascular disease/noncancer, and 1.29 (1.02–1.64) and 1.28 (1.03–1.60) for all causes in men and women, respectively. Long sleep duration of 10 hours or longer was associated with 1.5- to 2-fold increased mortality from total and ischemic stroke, total cardiovascular disease, noncardiovascular disease/noncancer, and all causes for men and women, compared with 7 hours of sleep in both sexes. There was no association between sleep duration and cancer mortality in either sex. Conclusions: Both short and long sleep duration were associated with increased mortality from cardiovascular disease, noncardiovascular disease/noncancer, and all causes for both sexes, yielding a U-shaped relationship with total mortality with a nadir at 7 hours of sleep. Citation: Ikehara S; Iso H; Date C; Kikuchi S; Watanabe Y; Wada Y; Inaba Y; Tamakoshi A. Association of sleep duration with mortality from cardiovascular disease and other causes for Japanese men and women: the JACC study. SLEEP 2009;32(3):259–301.